

**Mini Project 1: “Student Productivity Tracker” Objective: Combine loops, arrays, and file handling to build a console-based app that logs daily study hours and generates a weekly progress report. Trainer Coverage: Guide students to design menu-driven interface. Discuss file I/O for persistence.**

## **Tasks**

**1**

**Create a console-based app called Music Listening Logger that allows users to enter the number of minutes they listened to music each day for a week and stores this data in an array.**

**2.**

**Add a menu-driven interface to your Music Listening Logger so users can choose to log new listening minutes, view their weekly summary, or exit the app.<br><br><em><strong>Hint:</strong> Use a loop to repeatedly show the menu until the user selects exit.</em>**

**3.**

**Implement file handling in your Music Listening Logger so that when the user logs their daily listening minutes, the data is saved to a file named music\_log.txt for persistence.**

**4.**

**Add a feature to your app that reads the saved music\_log.txt file and generates a weekly report showing the total, average, and highest listening minutes in the week.**

**5.**

**Refactor your Music Listening Logger to allow users to reset their weekly data by clearing the array and deleting the contents of**

music\_log.txt.<br><br><em><strong>Constraint:</strong> Make sure the reset option is available in the menu and asks for confirmation before deleting data.</em>

### Code-

```
#include <stdio.h>
```

```
#include <stdlib.h>
```

```
#include <string.h>
```

```
#define DAYS 7
```

```
#define FILENAME "music_log.txt"
```

```
void showMenu(void);
```

```
void loadFromFile(int minutes[]);
```

```
void saveToFile(int minutes[]);
```

```
void logMinutes(int minutes[]);
```

```
void viewWeeklySummary(int minutes[]);
```

```
void resetData(int minutes[]);
```

```
void clearInputBuffer(void);
```

```
const char* dayName(int index);
```

```
int main(void) {
```

```
    int minutes[DAYS]
```

```
    int choice;
```

```
loadFromFile(minutes);
```

```
printf("=====\n");
```

```
printf(" Welcome to Music Listening Logger \n");
```

```
printf("=====\n");
```

```
int running = 1;
```

```
while (running) { /* loop until user selects Exit */
```

```
    showMenu();
```

```
    printf("Enter your choice: ");
```

```
    if (scanf("%d", &choice) != 1) {
```

```
        printf("Invalid input. Please enter a number.\n\n");
```

```
        clearInputBuffer();
```

```
        continue;
```

```
    }
```

```
    switch (choice) {
```

```
        case 1:
```

```
            logMinutes(minutes);
```

```
            break;
```

```
        case 2:
```

```
            viewWeeklySummary(minutes);
```

```
            break;
```

```
        case 3:
```

```
            resetData(minutes);
```

```

        break;

    case 4:

        printf("\nGoodbye! Keep enjoying your music. \n");

        running = 0;

        break;

    default:

        printf("\nInvalid choice. Please select 1-4.\n\n");

    }

}

return 0;

}

void showMenu(void) {

    printf("-----\n");

    printf("MENU\n");

    printf("1. Log listening minutes for a day\n");

    printf("2. View weekly summary/report\n");

    printf("3. Reset weekly data\n");

    printf("4. Exit\n");

    printf("-----\n");

}

const char* dayName(int index) {

    static const char *names[DAYS] = {

        "Monday", "Tuesday", "Wednesday", "Thursday",

        "Friday", "Saturday", "Sunday"
    }

```

```
};  
    return names[index];  
}
```

```
void clearInputBuffer(void) {  
    int c;  
    while ((c = getchar()) != '\n' && c != EOF) { /* discard */ }  
}
```

```
void loadFromFile(int minutes[]) {  
    FILE *fp = fopen(FILENAME, "r");  
  
    for (int i = 0; i < DAYS; i++) {  
        minutes[i] = -1;  
    }  
  
    if (fp == NULL) {  
        return;  
    }  
  
    for (int i = 0; i < DAYS; i++) {  
        if (fscanf(fp, "%d", &minutes[i]) != 1) {  
            minutes[i] = -1;  
        }  
    }  
}
```

```
}

fclose(fp);
}

void saveToFile(int minutes[]) {
    FILE *fp = fopen(FILENAME, "w");

    if (fp == NULL) {
        printf("Error: could not open %s for writing.\n", FILENAME);
        return;
    }

    for (int i = 0; i < DAYS; i++) {
        fprintf(fp, "%d\n", minutes[i]);
    }

    fclose(fp);
}

void logMinutes(int minutes[]) {
    int day, mins;

    printf("\nWhich day do you want to log?\n");
    for (int i = 0; i < DAYS; i++) {
        printf(" %d. %s\n", i + 1, dayName(i));
```

```
}

printf("Enter day number (1-%d): ", DAYS);

if (scanf("%d", &day) != 1 || day < 1 || day > DAYS) {
    printf("Invalid day number.\n\n");
    clearInputBuffer();
    return;
}

printf("Enter minutes listened on %s: ", dayName(day - 1));
if (scanf("%d", &mins) != 1 || mins < 0) {
    printf("Invalid number of minutes.\n\n");
    clearInputBuffer();
    return;
}

minutes[day - 1] = mins;
saveToFile(minutes);

printf("Logged %d minutes for %s. Saved to %s.\n\n",
        mins, dayName(day - 1), FILENAME);
}

void viewWeeklySummary(int minutes[]) {
    int total = 0, count = 0, highest = -1, highestDay = -1;
```

```

printf("\n===== Weekly Music Listening Report =====\n");

for (int i = 0; i < DAYS; i++) {
    if (minutes[i] >= 0) {
        printf(" %-10s: %d minutes\n", dayName(i), minutes[i]);
        total += minutes[i];
        count++;
        if (minutes[i] > highest) {
            highest = minutes[i];
            highestDay = i;
        }
    } else {
        printf(" %-10s: (not logged)\n", dayName(i));
    }
}

printf("-----\n");

if (count == 0) {
    printf("No data logged yet this week.\n");
} else {
    double average = (double) total / count;
    printf("Total minutes   : %d\n", total);
    printf("Average per day : %.2f minutes (over %d logged day%s)\n",
        average, count, count == 1 ? "" : "s");
    printf("Highest listening: %d minutes on %s\n",

```

```

        highest, dayName(highestDay));
    }
    printf("=====\n\n");
}

void resetData(int minutes[]) {
    char confirm;

    printf("\nAre you sure you want to reset all weekly data? (y/n): ");
    scanf(" %c", &confirm);
    clearInputBuffer();

    if (confirm == 'y' || confirm == 'Y') {
        /* Clear the array */
        for (int i = 0; i < DAYS; i++) {
            minutes[i] = -1;
        }

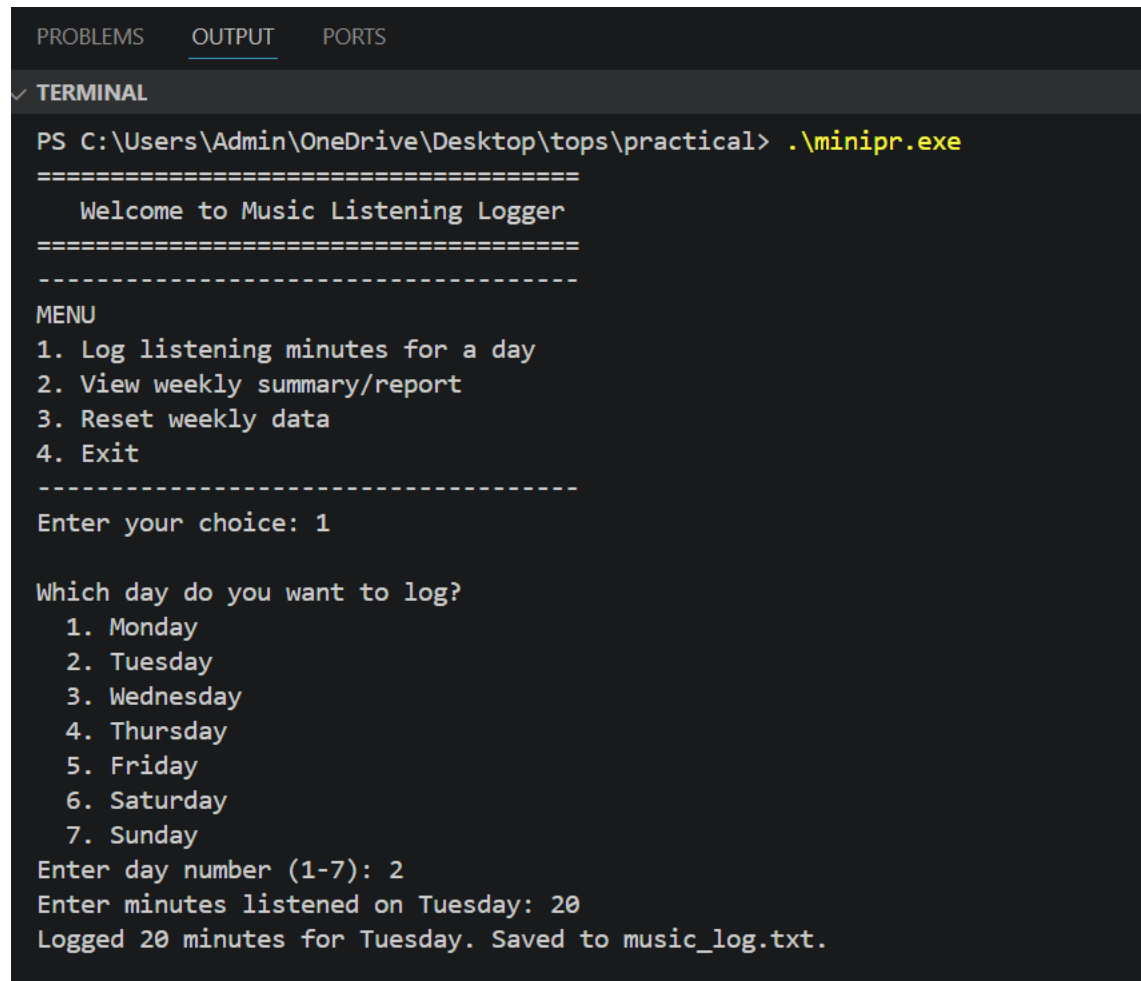
        FILE *fp = fopen(FILENAME, "w"); /* opening in "w" truncates it */
        if (fp != NULL) {
            fclose(fp);
        }
        remove(FILENAME);

        printf("All weekly data has been reset.\n\n");
    }
}

```

```
} else {  
    printf("Reset cancelled. Your data is safe.\n\n");  
}  
}
```

Output-



```
PROBLEMS  OUTPUT  PORTS  
✓ TERMINAL  
PS C:\Users\Admin\OneDrive\Desktop\tops\practical> .\minipr.exe  
=====  
Welcome to Music Listening Logger  
=====  
-----  
MENU  
1. Log listening minutes for a day  
2. View weekly summary/report  
3. Reset weekly data  
4. Exit  
-----  
Enter your choice: 1  
  
Which day do you want to log?  
1. Monday  
2. Tuesday  
3. Wednesday  
4. Thursday  
5. Friday  
6. Saturday  
7. Sunday  
Enter day number (1-7): 2  
Enter minutes listened on Tuesday: 20  
Logged 20 minutes for Tuesday. Saved to music_log.txt.
```

PROBLEMS OUTPUT PORTS

✓ TERMINAL

-----  
MENU

1. Log listening minutes for a day
  2. View weekly summary/report
  3. Reset weekly data
  4. Exit
- 

Enter your choice: 2

==== Weekly Music Listening Report ====

Monday : (not logged)  
Tuesday : 20 minutes  
Wednesday : (not logged)  
Thursday : (not logged)  
Friday : (not logged)  
Saturday : (not logged)  
Sunday : (not logged)

-----  
Total minutes : 20  
Average per day : 20.00 minutes (over 1 logged day)  
Highest listening: 20 minutes on Tuesday

=====

=====

-----  
MENU

1. Log listening minutes for a day
  2. View weekly summary/report
  3. Reset weekly data
  4. Exit
- 

Enter your choice: 3

Are you sure you want to reset all weekly data? (y/n): 4  
Reset cancelled. Your data is safe.

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